

Lesson 4 Basic Ros Case 1- Creating messages and services

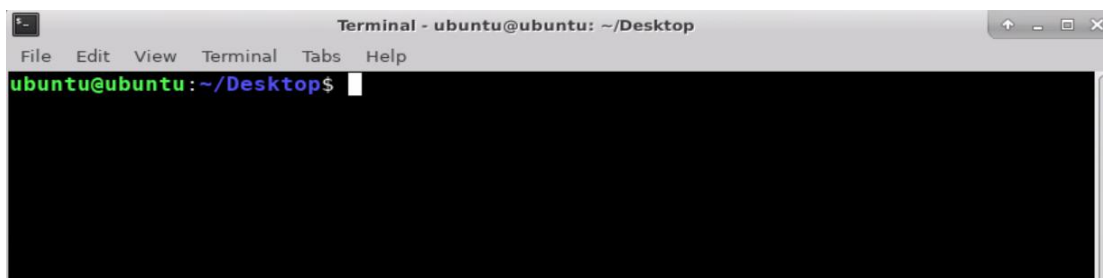
1. Message and service

msg (message): msg is text file used to describe the data value published by the ROS node. It can generate source code for messages written in different programming languages.

srv (service): srv is used to describe the Ros service type. It, consisting of request and response, directly builds on the Ros msg format to achieve request and response communication between nodes.

2. Create a Message

1) We need to define a new message in the software package created before. Initially, please open remote desktop of NoMachine, and right-click and select “**Open Teminal Here**” to open Linux terminal.



1) Enter “**roscd beginner_hiwonder**” command and press “Enter” to access the software package path.



● Attention: If “**No such package/stack ‘beginner_hiwonder’**” occurs, software package may be not stored in environment variable

ROS_PACKAGE_PATH.

```
ubuntu@ubuntu:~/Desktop$ roscd beginner_hiwonder
roscd: No such package/stack 'beginner_hiwonder'
ubuntu@ubuntu:~/Desktop$
```

And we need to add the path of the software package to the environment variable **ROS_PACKAGE_PATH**.

1) Firstly, enter **"echo "source ~/catkin_ws/devel/setup.bash" >> ~/.bashrc"** command and press "Enter".

```
ubuntu@ubuntu:~/Desktop$ echo "source ~/catkin_ws/devel/setup.
bash" >> ~/.bashrc
ubuntu@ubuntu:~/Desktop$
```

2) Then, enter **"source ~/.bashrc"** command and press "Enter" to complete.

```
ubuntu@ubuntu:~/Desktop$ source ~/.bashrc
ubuntu@ubuntu:~/Desktop$
```

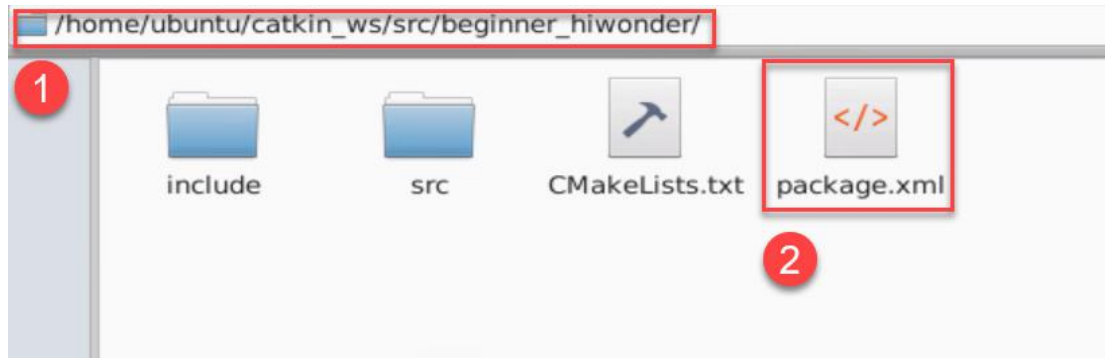
3) Enter **"mkdir msg"** command and press "Enter" to create a msg folder.

```
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder$ mkdir msg
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder$
```

4) Enter **"echo "int64 num" > msg/Num.msg"** command and press "Enter" to write int64 num message field to the Num.msg file.

```
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder$ echo "int64 num" > msg/Num.msg
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder$
```

5) Open the package.xml saved in the path of **"/home/ubuntu/catkin_ws/src/beginner_hiwonder/"**, and ensure it contains following two lines, framed in the picture, and not be commented. If not contained, the following two lines should be added.



```
<buildtool_depend>catkin</buildtool_depend>
<build_depend>roscpp</build_depend>
<build_depend>rospy</build_depend>
<build_depend>std_msgs</build_depend>
<build_export_depend>roscpp</build_export_depend>
<build_export_depend>rospy</build_export_depend>
<build_export_depend>std_msgs</build_export_depend>
<exec_depend>roscpp</exec_depend>
<exec_depend>rospy</exec_depend>
<exec_depend>std_msgs</exec_depend>
<build_depend>message_generation</build_depend>
<exec_depend>message_runtime</exec_depend>

<!-- The export tag contains other, unspecified, tags -->
<export>
  <!-- Other tools can request additional information be placed here -->
</export>
</package>
```

6) Open CMakeLists.txt saved in the path of “/home/ubuntu/catkin_ws/src/beginner_hiwonder/”, and call and add a **message_Generation** dependency to an existing find_package.

```
find_package(catkin REQUIRED COMPONENTS
  roscpp
  rospy
  std_msgs
  message_generation
)
```

7) In the same file, add the. **MSG** file manually. Find the corresponding program block and modify according to the following figure.

```
## Generate messages in the 'msg' folder
#add_message_files(
#  FILES
#  Message1.msg
#  Message2.msg
#)
```

```
## Generate messages in the 'msg' folder
add_message_files(
  FILES
  Num.msg
)
```

8) To ensure **generate_messages()** is called, please modify in the same file according to the following figure

```
## Generate added messages and services with any dependencies listed here
# generate_messages(
#   DEPENDENCIES
#   std_msgs
#)
```

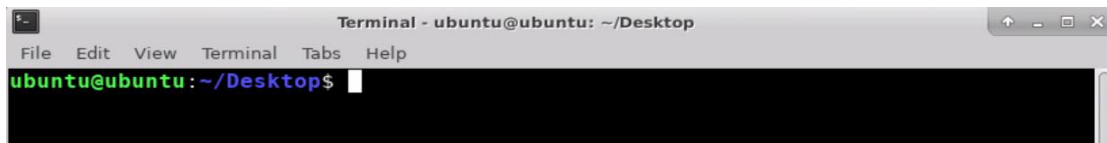
9) Enter “**rosmmsg show beginner_hiwonder/Num**” command and press “Enter” to check whether the written message field is recognized by Ros system.

```
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder$ rosmmsg show beginner_hiwonder/Num
int64 num
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder$
```

3. Create a service

1) Based on the software package created before, we need to create a

service. Please open remote desktop of NoMachine, and right-click and select “Open Terminal Here” to open Linux terminal.



```
Terminal - ubuntu@ubuntu: ~/Desktop
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~/Desktop$
```

2) Enter “**roscd beginner_hiwonder**” command and press “Enter” to access the software package path.



```
ubuntu@ubuntu:~/Desktop$ roscd beginner_hiwonder
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder$
```

3) Enter “**mkdir srv**” command and press “Enter” to create a srv folder.



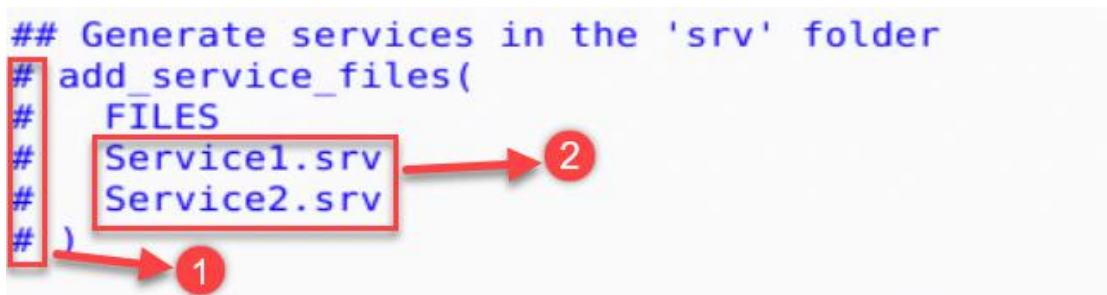
```
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder$ mkdir srv
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder$
```

4) Enter “**roscp rospy_tutorials AddTwoInts.srv srv/AddTwoInts.srv**” command and press “Enter”. Copy **AddTwoInts** in rospy_tutorials and paste it to beginner_hiwonder.



```
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder$ roscp rospy_
tutorials AddTwoInts.srv srv/AddTwoInts.srv
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder$
```

5) Open CMakeLists.txt saved in the path of **/home/ubuntu/catkin_ws/src/beginner_hiwonder/** and add .srv file manually. Then modify the corresponding program block as follows.



```
## Generate services in the 'srv' folder
# add_service_files(
#   FILES
#   Service1.srv
#   Service2.srv
# )
```

```
## Generate services in the 'srv' folder
add_service_files(
  FILES
  AddTwoInts.srv 3
)
```

6) Enter “**rossrv show beginner_hiwonder/AddTwoInts**” command and press “Enter” to check whether the written message field is recognized by Ros system.

```
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder$ rossrv show beginner_hiwonder
/AddTwoInts
int64 a
int64 b
---
int64 sum
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder$
```